

Three-dimensional non-free-parameter lattice-Boltzmann model and its application to inviscid compressible flows

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ABSTRACT

In this Letter, a three-dimensional (3D) lattice-Boltzmann model is presented following the non-free-parameter lattice-Boltzmann method of Qu et al. [K. Qu, C. Shu, Y.T. Chew, Phys. Rev. E 75 (2007) 036706]. A simple function, which satisfies the zeroth- through third-order moments of the Maxwellian distribution function, is introduced to replace the Maxwellian distribution function as the continuous equilibrium distribution function in 3D space. The function is then discretized to discrete-velocity directions via a 25-point Lagrangian interpolation polynomial. To simulate compressible flows with shock waves, an implicit-explicit finite-difference scheme based on the total variation diminishing flux limitation is adopted to solve the discrete Boltzmann-BGK equation in order to capture the shock waves in compressible flows with a finite number of grid points. The model is validated by its application to some typical inviscid compressible flows ranging from 1D to 3D, and the numerical results are found to be in excellent agreement with the analytical solutions and/or other numerical results.

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1. Introduction

In the past two decades, the lattice-Boltzmann (LB) method has been developed into an alternative numerical method to simulate fluid flows and model physics in fluids [1–4]. In the LB method, instead of solving the macroscopic governing equations, the discrete Boltzmann equation is solved to simulate the flow of a Newtonian fluid with collision models such as the Bhatnagar–Gross–Krook (BGK) model [5,6]. Unlike conventional numerical methods, which are based on the discretization of macroscopic governing equations, and unlike the molecular dynamics (MD) method, which is based on molecular representation with complicated molecule collision rules, the LB method is based on the mesoscopic kinetic equation for particle distribution functions.

Due to some inherent features of the LB method, such as the simple form of the governing equation, the avoidance of nonlinear convective terms, the inherent parallelizability on multiple processors [7], and the effectiveness for time-dependent fluid computations, the LB method is a very attractive representation of the nonlinear macroscopic systems. Meanwhile, the solution of the nonlinear macroscopic systems, especially for those containing

shock waves or contact discontinuities, is a very interesting work in computational fluid dynamics (CFD).

In the past years, many efforts have been made in constructing LB models for the compressible Euler equations. The first two-dimensional (2D) LB model for the compressible Euler equations was proposed by Hu et al. [8] based on a triangular lattice. This model was established under the assumption that the particles moving along every link can be separated into two kinds, A and B , with different energy levels ε_A and ε_B , and the rest particles possess energy level ε_D . Similar to Hu et al.'s model, Yan et al. [9] presented a 2D LB model with three energy levels on a square lattice and they pointed out that the LB method would become a very interesting method when it is combined with the high resolution techniques, such as the total variation diminishing (TVD) scheme and the essentially non-oscillatory (ENO) scheme. Shi et al. [10] also constructed a 2D LB model for the compressible Euler equations. Instead of using the standard LB scheme (the streaming-collision process), they employed the finite-difference LB method. To improve the solution accuracy, the TVD scheme was adopted to solve the discrete Boltzmann-BGK equation.

In the literature, Kataoka et al. [11] has presented the theoretical background of the LB method for the compressible Euler equations. The important finding in their study implies that discontinuity-capturing schemes should be applied to solve the discrete Boltzmann-BGK equation in order to capture the shock

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waves in compressible flows with a finite number of grid points. Based on this finding, they developed several models for the compressible Euler equations. However, these models together with the above-mentioned models often include many free parameters which should be carefully chosen to get satisfied simulation results. But how to tune these parameters is not clear. On the other hand, only the results of subsonic cases with weak shock waves were presented in these studies.

To enable the simulation of high speed flows, Sun et al. [12–14] developed a locally adaptive LB method. In this method, a simplified equilibrium distribution function, the Kronecker delta function, is adopted to replace the Maxwellian distribution function as the continuous equilibrium distribution function; the pattern of lattice velocities varies with the local flow velocity and the internal energy. The locally adaptive LB method permits the mean flow to have high Mach number. However, since the lattice velocity pattern varies with the local flow velocity, then the particles are no longer simply hopping from one grid point to the next. A reconstruction step is therefore required to obtain the values on the grid-nodes and interpolations have to be used, which will introduce undesirable numerical artifacts [15].

Recently, based on the idea of simplified equilibrium distribution functions of the locally adaptive LB method, Qu et al. [16] developed a non-free-parameter LB method to construct equilibrium distribution functions. A circular function was introduced to replace the Maxwellian distribution function for 2D problems and a Lagrangian interpolation polynomial was then constructed to discretize the circular function to the discrete velocity directions. Different from the adaptive LB method, in this method the lattice velocity pattern is fixed. In addition, the equilibrium distribution function and the lattice velocity set are derived without assuming specific forms, which makes the equilibrium distribution containing no free parameters. The main features of this method can be summarized as follows: (1) the low-Mach number limit is breached in discrete equilibrium distribution functions; (2) no special treatment is required for the isotropy of lattice tensors; and (3) there are no free parameters in this method.

In this Letter, following the non-free-parameter LB method, a 3D non-free-parameter LB model is presented. A simple function, which satisfies the zeroth- through third-order moments of the Maxwellian distribution function, is used to replace the Maxwellian distribution function as the continuous equilibrium distribution function in 3D space. The function is then discretized to discrete-velocity directions via a 25-point Lagrangian interpolation polynomial. Through the Chapman–Enskog analysis, the compressible Navier–Stokes equations and the energy equation at the Euler level with a specific-heat ratio, $\gamma = (D + 2)/D$, can be recovered. To enable the simulation of a realistic compressible flow, the specific-heat ratio should be made adjustable. A recently developed approach to adjust the specific-heat ratio [17] is therefore introduced into the non-free-parameter LB models. The model is validated by its applications to some typical inviscid compressible flows ranging from 1D to 3D. In the computation, like the earlier work [10,16], the finite-difference scheme based on the TVD flux limitation is used to solve the discrete Boltzmann–BGK equation in order to capture the shock waves with a finite number of grid points. In particular, to improve the numerical stability, an implicit–explicit (IMEX) Runge–Kutta scheme is adopted for time discretization. It is because of the high performance of the IMEX Runge–Kutta schemes that a relatively large time step can be chosen.

2. Model description

The non-free-parameter LB method can be briefly summarized as follows: firstly, a simple function, which satisfies the needed

statistical relations, is devised to replace the Maxwellian distribution function as the continuous equilibrium distribution function; secondly, a Lagrangian interpolation polynomial is constructed as an assigning function to discretize the simple function to the discrete velocity directions. In Ref. [16], the following circular function was presented to replace the Maxwellian distribution function for 2D problems:

$$f^{eq} = \begin{cases} \frac{\rho}{2\pi r} & \text{if } \|\xi - \mathbf{u}\| = \|\mathbf{r}\| = r = \sqrt{2RT}, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where $\mathbf{r} = \xi - \mathbf{u}$ is the effective peculiar velocity and r is its corresponding module. Eq. (1) means that all the particles are concentrated on a circle. The integral over the microscopic velocity space is reduced to the one along the arclength of the circle: $d\xi = r d\vartheta$, where ϑ is the angle of the position vector on the circle with the x axis.

Similar to the Maxwellian distribution function, the circular function is also continuous in the phase space and cannot be directly applied in the LB method. By noticing that the approach to assign a variable from one point to several other points is widely used in particle methods, such as the particle-in-cell method, vortex method, and vortex-in-cell method, Qu et al. introduced an assigning function to discretize the circular function onto a lattice. For any $d\rho$ along the circle, it has a contribution $\phi_i(\xi) d\rho$ in the lattice direction $\mathbf{e}_i$, where $\phi_i(\xi)$ is the so-called assigning function. The contribution of the whole circular function to $\mathbf{e}_i$ direction is given by

$$\rho_i = f_i^{eq} = \int \phi_i(\xi) d\rho = \int \phi_i(\xi) f^{eq} d\xi, \quad (2)$$

where f_i^{eq} is the discrete equilibrium density distribution function. To satisfy the following velocity moments' condition:

$$\sum_i f_i^{eq} = \rho, \quad \sum_i f_i^{eq} \mathbf{e}_i = \rho \mathbf{u}, \quad (3a)$$

$$\sum_i f_i^{eq} e_{i\alpha} e_{i\beta} = \rho u_\alpha u_\beta + p \delta_{\alpha\beta}, \quad (3b)$$

$$\sum_i f_i^{eq} e_{i\alpha} e_{i\beta} e_{i\gamma} = \rho u_\alpha u_\beta u_\gamma + p(u_\gamma \delta_{\alpha\beta} + u_\beta \delta_{\alpha\gamma} + u_\alpha \delta_{\beta\gamma}), \quad (3c)$$

$$\sum_i f_i^{eq} \frac{1}{2} e_i^2 = \frac{1}{2} \rho (u^2 + DRT),$$

$$\sum_i f_i^{eq} \frac{1}{2} e_i^2 e_{i\alpha} = \frac{1}{2} \rho [u^2 + (D + 2)RT] u_\alpha, \quad (3d)$$

where $p = \rho RT$ (R is the specific gas constant) is the pressure, D is the spatial dimension, $e_i^2 = \mathbf{e}_i \cdot \mathbf{e}_i$, and $\delta_{\alpha\beta}$, $\delta_{\alpha\gamma}$, and $\delta_{\beta\gamma}$ are the Kronecker delta with two indices, the assigning function $\phi_i(\xi)$ should be constraint by [16]

$$\sum_i \phi_i(\xi) = 1, \quad \sum_i z_{i\alpha} \phi_i(\xi) = 0, \quad (4a)$$

$$\sum_i z_{i\alpha} z_{i\beta} \phi_i(\xi) = 0, \quad \sum_i z_{i\alpha} z_{i\beta} z_{i\gamma} \phi_i(\xi) = 0, \quad (4b)$$

where $z_{i\alpha}$ is given by $z_{i\alpha} = e_{i\alpha} - \xi_\alpha$. It has been found [16,18] that these constraints can be satisfied by taking a third-order Lagrangian interpolation polynomial as the assigning function.

The circular function was devised for 2D problems. For 3D inviscid compressible flows, the following function can be introduced to replace the Maxwellian distribution function:

$$g^{eq} = \begin{cases} \frac{\rho}{4\pi r^2} & \text{if } \|\xi - \mathbf{u}\| = \|\mathbf{r}\| = r, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

The above equation means that all the particles are concentrated on a spherical surface located in a 3D space of ξ . The integral over

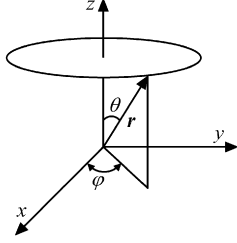


Fig. 1. Sketch of the angles θ and φ .

the microscopic velocity space is then reduced to the one upon the spherical surface: $d\xi = r^2 \sin\theta d\theta d\varphi$, where the angles θ and φ are shown in Fig. 1. After some algebra, the following results can be obtained:

$$\int g^{eq} d\xi = \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} r^2 \sin\theta d\theta d\varphi \equiv \rho, \quad (6a)$$

$$\int g^{eq} \xi_\alpha d\xi = \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} (\mathbf{u} + \mathbf{r})_\alpha r^2 \sin\theta d\theta d\varphi \equiv \rho u_\alpha, \quad (6b)$$

$$\begin{aligned} \int g^{eq} \xi_\alpha \xi_\beta d\xi &= \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} (\mathbf{u} + \mathbf{r})_\alpha (\mathbf{u} + \mathbf{r})_\beta r^2 \sin\theta d\theta d\varphi \\ &= \rho u_\alpha u_\beta + \frac{1}{3} \rho r^2 \delta_{\alpha\beta}, \end{aligned} \quad (6c)$$

$$\begin{aligned} \int g^{eq} \xi_\alpha \xi_\beta \xi_\gamma d\xi &= \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} \xi_\alpha \xi_\beta \xi_\gamma r^2 \sin\theta d\theta d\varphi \\ &= \rho u_\alpha u_\beta u_\gamma + \frac{1}{3} \rho r^2 (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}), \end{aligned} \quad (6d)$$

$$\begin{aligned} \int g^{eq} \frac{1}{2} \xi^2 d\xi &= \frac{1}{2} \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} (\mathbf{u} + \mathbf{r}) \cdot (\mathbf{u} + \mathbf{r}) r^2 \sin\theta d\theta d\varphi \\ &= \frac{1}{2} (\rho u^2 + \rho r^2), \end{aligned} \quad (6e)$$

$$\begin{aligned} \int g^{eq} \frac{1}{2} \xi^2 \xi_\alpha d\xi &= \frac{1}{2} \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi r^2} (u^2 + 2\mathbf{u} \cdot \mathbf{r} + r^2) (\mathbf{u} + \mathbf{r})_\alpha r^2 \sin\theta d\theta d\varphi \\ &= \frac{1}{2} (\rho u^2 + \rho r^2) u_\alpha + \frac{1}{3} \rho r^2 u_\alpha. \end{aligned} \quad (6f)$$

In order to satisfy the zeroth- through third-order moments of the Maxwellian distribution function in 3D space, r^2 should be identical with $3RT$. Thus r in Eq. (5) should be defined as $r = \sqrt{3RT}$.

To discretize g^{eq} onto a lattice, a 3D Lagrangian interpolation polynomial is needed. In this study, the following 25-point polynomial is used [19]:

$$\begin{aligned} P(x, y, z) &= a_0 + a_1 x + a_2 y + a_3 z + a_4 x^2 + a_5 y^2 + a_6 z^2 \\ &\quad + a_7 xy + a_8 xz + a_9 yz + a_{10} x^3 + a_{11} y^3 + a_{12} z^3 \\ &\quad + a_{13} x^2 y + a_{14} x^2 z + a_{15} y^2 z + a_{16} xy^2 + a_{17} xz^2 \\ &\quad + a_{18} z^2 y + a_{19} x^4 + a_{20} y^4 + a_{21} z^4 + a_{22} x^2 y^2 \\ &\quad + a_{23} x^2 z^2 + a_{24} y^2 z^2. \end{aligned} \quad (7)$$

The assigning function can be written as $\phi_i(x, y, z) = \mathbf{a}_i \mathbf{t}$, where $\mathbf{a}_i = [a_{i,0}, a_{i,1}, a_{i,2}, \dots, a_{i,24}]$, $\mathbf{t} = [1, x, y, \dots, y^2 z^2]$. Meanwhile, a 3D 25-velocity (D3Q25) lattice is adopted

$$\mathbf{e}_i = \begin{cases} (0, 0, 0), & i = 0, \\ (\pm 1, 0, 0)c, (0, \pm 1, 0)c, (0, 0, \pm 1)c, \\ & i = 1, 2, \dots, 6, \\ (\pm 1, \pm 1, 0)c, (\pm 1, 0, \pm 1)c, (0, \pm 1, \pm 1)c, \\ & i = 7, 8, \dots, 18, \\ (\pm 2, 0, 0)c, (0, \pm 2, 0)c, (0, 0, \pm 2)c, \\ & i = 19, 20, \dots, 24, \end{cases} \quad (8)$$

where $c = \sqrt{3RT_c}$ is the lattice speed of the D3Q25 lattice. Since the assigning function $\phi_i(x, y, z)$ is a Lagrangian interpolation polynomial, then it has the δ property: $\phi_i(x_j, y_j, z_j) = \delta_{ij}$, where $i, j = 0, 1, \dots, 24$ and (x_j, y_j, z_j) are the x , y , and z components of the lattice velocity along the j th direction. According to this property, the assigning function, $\phi_i(x, y, z)$, $i = 0, 1, 7, 19$, can be obtained as follows:

$$\begin{aligned} \phi_0(x, y, z) &= 1 + x^2 y^2 + x^2 z^2 + y^2 z^2 - \frac{5}{4} x^2 - \frac{5}{4} y^2 - \frac{5}{4} z^2 \\ &\quad + \frac{1}{4} x^4 + \frac{1}{4} y^4 + \frac{1}{4} z^4, \\ \phi_1(x, y, z) &= \frac{2}{3} x - \frac{1}{2} xy^2 - \frac{1}{2} xz^2 - \frac{1}{2} x^2 y^2 - \frac{1}{2} x^2 z^2 + \frac{2}{3} x^2 \\ &\quad - \frac{1}{6} x^3 - \frac{1}{6} x^4, \\ \phi_7(x, y, z) &= \frac{1}{4} (xy + x^2 y + xy^2 + x^2 y^2), \\ \phi_{19}(x, y, z) &= \frac{1}{24} (-2x - x^2 + 2x^3 + x^4). \end{aligned} \quad (9)$$

The other assigning functions can be obtained according to the symmetry of the D3Q25 lattice. For instance $\phi_3(x, y, z)$ can be calculated with the same form of $\phi_1(x, y, z)$, but the locations of x and y in $\phi_1(x, y, z)$ should be permuted, namely $\phi_3(x, y, z) = \phi_1(y, x, z)$. The discrete equilibrium density distribution function is then obtained by (see Appendix A for details)

$$f_i^{eq} = \int_0^{2\pi} \int_0^\pi \frac{\rho}{4\pi} \phi_i(\xi_x, \xi_y, \xi_z) \sin\theta d\theta d\varphi, \quad (10)$$

where ξ_x , ξ_y , and ξ_z are given by $u_x + r \sin\theta \cos\varphi$, $u_y + r \sin\theta \sin\varphi$, and $u_z + r \cos\theta$, respectively.

In what follows the computational scheme for simulating incompressible flows will be presented. We start with the following discrete Boltzmann-BGK equation:

$$\partial_t f_i + (\mathbf{e}_i \cdot \nabla) f_i = -\frac{1}{\tau} (f_i - f_i^{eq}), \quad (11)$$

where f_i is the discrete distribution function and τ is the relaxation time. If the macroscopic temperature is calculated through $\rho(u^2 + DRT)/2 = \sum_i f_i e_i^2 / 2$, then the following macroscopic equations can be obtained according to Eqs. (3) (see Appendix B for details):

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (12a)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \Pi + O(K^2), \quad (12b)$$

$$\partial_t (\rho E) + \nabla \cdot [(\rho E + p) \mathbf{u}] = 0 + O(K), \quad (12c)$$

where $\Pi = \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T - (2/D)(\nabla \cdot \mathbf{u}) \mathbf{I}]$ is the viscous stress tensor, $\mu = \tau p$ is the shear viscosity, K , which is proportional to the Knudsen number, is the expansion parameter used in the

Chapman–Enskog procedure and $E = (u^2 + DRT)/2$ is the total energy including only D degrees of freedom. As a result, the specific-heat ratio is given by $\gamma = (D + 2)/D$. For 2D and 3D models, γ is fixed at 2 and 5/3, respectively.

When the viscous term in Eq. (12b) is regarded as the discretization error, Eqs. (12) become an inviscid Euler system. However, to simulate a realistic inviscid compressible flow, the specific-heat ratio should be made adjustable. Recently, Nie et al. [17] proposed a general approach for treating the effect of the internal degrees of freedom of the polyatomic gas molecules to the equations of conservative quantities. As one direct application of the result, the specific-heat ratio can be made adjustable. In this approach, the mesoscopic kinetic equation is given by [17]

$$\partial_t f_{\alpha,n} + (\mathbf{e}_\alpha \cdot \nabla) f_{\alpha,n} = -\frac{1}{\tau} (f_{\alpha,n} - h_n f_{\alpha,n}^{eq}), \quad n = 1, 2, \quad (13)$$

where $f_{\alpha,1}$ and $f_{\alpha,2}$ are two reduced distribution functions; $h_1 = 1$ and $h_2 = RT$. The macroscopic variables are calculated by

$$\begin{aligned} \rho &= \sum_{\alpha} f_{\alpha,1}, \quad \rho \mathbf{u} = \sum_{\alpha} f_{\alpha,1} \mathbf{e}_\alpha, \\ \rho \tilde{E} &= \frac{1}{2} \rho (u^2 + bRT) = \frac{1}{2} \sum_{\alpha} f_{\alpha,1} e_{\alpha}^2 + \frac{S}{2} \sum_{\alpha} f_{\alpha,2}, \end{aligned} \quad (14)$$

where $\tilde{E} = (u^2 + bRT)/2$ is the total energy including b degrees of freedom and $S = b - D$. Through the Chapman–Enskog analysis, the macroscopic energy equation at the Euler level is now given as follows:

$$\partial_t (\rho \tilde{E}) + \nabla \cdot [(\rho \tilde{E} + p) \mathbf{u}] = 0 + O(K). \quad (15)$$

The specific-heat ratio is now given by $\gamma = (b + 2)/b$, which can be adjusted by tuning S .

To solve Eq. (13) numerically for compressible flows with shock waves, like the earlier work [10,16], the finite-difference-based LB scheme with the TVD flux limitation is used in order to capture the shock waves in compressible flows with a finite number of grid points. But for the time discretization, we adopt the implicit–explicit (IMEX) Runge–Kutta schemes [20], which consist of an implicit time discretization for the relaxation term and an explicit one for the other terms. It is because of the high performance of the IMEX Runge–Kutta schemes that we can choose a relatively large time step and the accuracy in time can be improved.

Although an implicit scheme is included in the IMEX Runge–Kutta scheme, its implicitness can be completely eliminated in the LB method according to the characteristic of the collision invariants, and then the discretized equation for the distribution function is still explicit. Details about this point can be found in Refs. [21–23]. Taking the first-order IMEX Runge–Kutta scheme for example, the following computation is conducted to advance $f_{\alpha,n}$ from time t to $t + \Delta t$:

$$f_{\alpha,n}^{t+\Delta t} = f_{\alpha,n}^t - \Delta t (\mathbf{e}_\alpha \cdot \nabla f_{\alpha,n}^{(1)}) + \frac{\Delta t}{\tau} (h_n^t f_{\alpha,n}^{eq,t} - f_{\alpha,n}^{(1)}), \quad (16)$$

where $f_{\alpha,n}^{(1)}$ is given by:

$$f_{\alpha,n}^{(1)} = \frac{f_{\alpha,n}^t + (\Delta t/\tau) h_n^t f_{\alpha,n}^{eq,t}}{1 + (\Delta t/\tau)}. \quad (17)$$

The advantages of implicit schemes are retained and their shortcomings can be overcome simultaneously. As a result, the numerical stability and the computational convergence rate are improved compared with other finite-difference-based LB schemes.

Note that the discretization of the spatial gradient operator ∇ can be implemented in a general body-fitted curvilinear coordinate ξ with non-uniform grids. For such a coordinate, the advection term $\mathbf{e}_\alpha \cdot \nabla f_\alpha$ can be written as

$$\mathbf{e}_\alpha \cdot \nabla f_\alpha = e_{\alpha i} \frac{\partial f_\alpha}{\partial r_i} = c_{\alpha j} \frac{\partial f_\alpha}{\partial \xi_j}, \quad (18)$$

where $c_{\alpha j} = e_{\alpha i} \partial \xi_j / \partial r_i$. More details about implementing finite-difference LB method on body-fitted curvilinear coordinates with non-uniform grids can be found in Refs. [23,25–27].

Besides for simulating inviscid compressible flows, the non-free-parameter LB models can also be applied to simulate isothermal viscous flows. When the temperature T in Eq. (A.1) is taken as a constant temperature T_0 , the following macroscopic equations can be obtained through the Chapman–Enskog analysis:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (19a)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \Pi' + O(K^2), \quad (19b)$$

where $\Pi' = \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$ and μ is now given by $\mu = \tau \rho RT_0$. It can be seen that Eq. (19b) does not contain the third-order deviation term which appears in the momentum equation recovered from the standard LB models (D2Q9, D3Q15, B3Q19, and D3Q27). In the literature, it has been demonstrated that this deviation term will reduce the accuracy of the LB method, for instance, in the initial stage of 2D decaying turbulence [24]. For simulating isothermal flows, the standard LB scheme (the streaming–collision process) is enough and the approach to adjust the specific-heat ratio is not needed.

3. Numerical tests

In this section, to validate the proposed model, numerical simulations are carried out for some typical compressible flows ranging from 1D to 3D. In simulations, the reference density ρ_0 and the reference temperature T_0 are used, and then the reference velocity and the reference pressure are defined as $u_0 = \sqrt{RT_0}$ and $p_0 = \rho_0 RT_0$, respectively. Unless otherwise mentioned, the first-order IMEX Runge–Kutta scheme is used and we set $\rho_0 = 1.165 \text{ kg/m}^3$, $T_0 = 303 \text{ K}$, $R = 287 \text{ J/(kg K)}$, $\gamma = 1.4$ ($S = 2$), $\mu = 1.86 \times 10^{-5} \text{ kg/(m s)}$ (air under normal condition), and $\tau_0 = \mu/p_0$. The characteristic temperature T_c is often set to be a value which is a little bit larger than the maximum stagnation temperature in the whole flow field, $T_c \geq \max(T^*)$ where $T^* = T[1 + (\gamma - 1)Ma^2/2]$.

1D test: Riemann problem

The flow of the Riemann problem includes a shock wave, a contact surface, and an expansion wave; hence, it makes a wonderful model problem on which to study the performance of the numerical schemes in the simulation of compressible flows. The well-known 1D Riemann problem with the following initial condition is considered:

$$\begin{aligned} (\rho/\rho_0, u_x/u_0, p/p_0) &= (1, 0, 1), \quad 0 < x/L_0 \leq 1/2, \\ (\rho/\rho_0, u_x/u_0, p/p_0) &= (0.125, 0, 0.1), \quad 1/2 < x/L_0 \leq 1. \end{aligned} \quad (20)$$

Here $L_0 = 1 \text{ m}$ is the reference length. The characteristic time of the system is defined as $t_0 = L_0/u_0$. In our simulation, a uniform mesh $N_x \times N_y \times N_z = 400 \times 2 \times 2$ is employed, where N_x , N_y , and N_z are the lattice numbers along the x -, y -, and z -directions, respectively. In the x -direction, $f_{\alpha,n} = f_{\alpha,n}^{eq}$ is set, and the periodic boundary conditions are imposed in the y - and z -directions.

The profiles of the density, velocity, pressure, and temperature are shown in Fig. 2, where the analytical solutions are presented with solid lines for comparison. In this test, T_c is set to be $4T_0/3$ and the time step Δt is taken as $7500\tau_0$. The numerical results are found to be in good agreement with the analytical ones.

2D test: Regular shock reflection

A steady 2D compressible flow, a regular shock reflection on a wall, is considered in this test. This problem involves three flow regions separated by an oblique shock and its reflection from a wall.

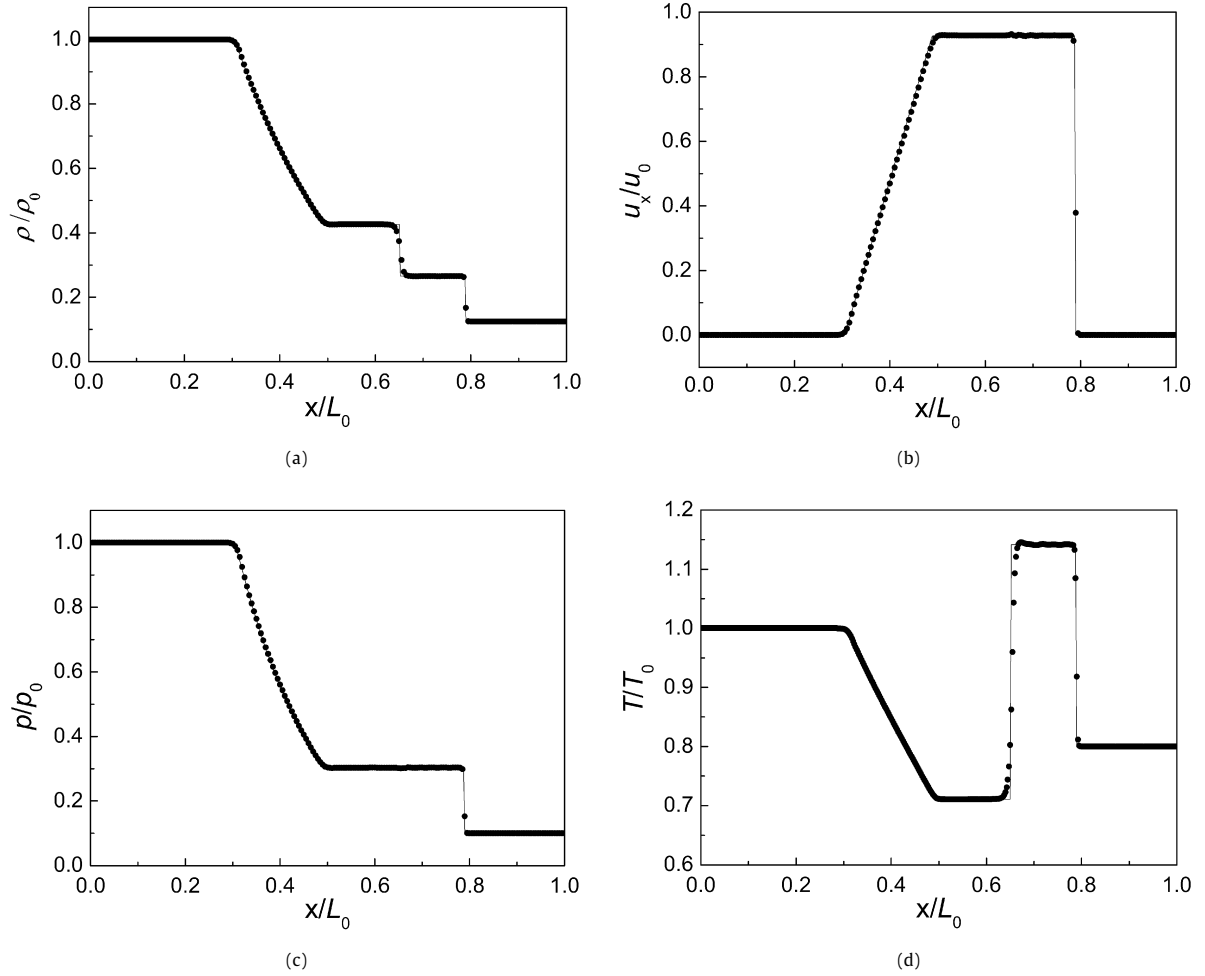


Fig. 2. Simulation of Riemann problem: comparisons between the numerical and analytical solutions at $t = 0.1644t_0$: (a) density, (b) velocity, (c) pressure, and (d) temperature.

The incoming shock wave of Mach number 2.9 has an incident angle of 29° to the wall. The Dirichlet conditions

$$\begin{aligned}
 (\rho/\rho_0, u_x/u_0, u_y/u_0, u_z/u_0, p/p_0)|_L &= (1.0, 2.9, 0, 0, 1/1.4), \\
 (\rho/\rho_0, u_x/u_0, u_y/u_0, u_z/u_0, p/p_0)|_T &= (1.69997, 2.61934, -0.50633, 0, 1.52819),
 \end{aligned}$$

are imposed on the left and top boundaries, respectively [12]. The bottom boundary is a reflecting wall. On this reflective wall, two levels of mirror points are used and the specular boundary condition is applied to consider the velocity slip at the wall. The right boundary is supersonic outflow where the zeroth-order extrapolation is used. The periodic boundary condition is applied in the z -direction. Initially, the solution of the entire domain is set to be that at the left boundary. In the computation, a uniform mesh $N_x \times N_y \times N_z = 200 \times 100 \times 2$ is employed. The characteristic temperature T_c is set to be $2T_0$ and $\Delta t = 20000\tau_0$. Fig. 3 shows the density and pressure contours in which the shock reflection is well captured. The incident angle measured according to the numerical results is 29.05° [$\arctan(1/1.8)$], which is accurate compared with the exact result 29° .

3D tests: Explosion in a box and expanding shock in open space

In this subsection, two 3D tests are considered: explosion in a box [28] and expanding shock in open space [29] as shown in Figs. 4(a) and 4(b), respectively. Namely, a spherical shock-wave expands in an enclosed box and in open space, respec-

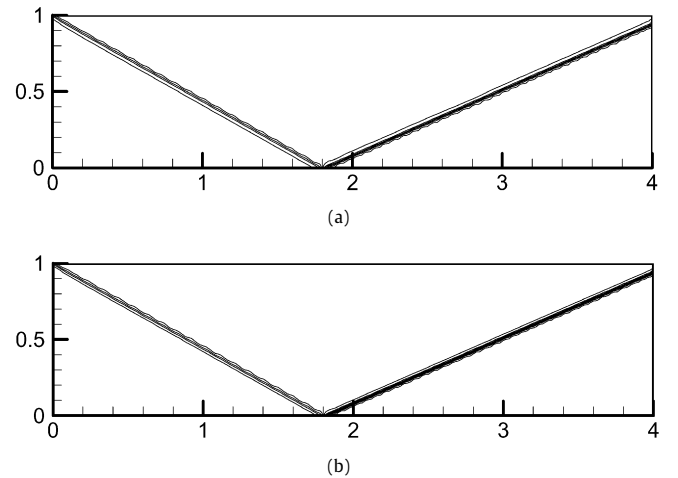


Fig. 3. Density (a) and pressure (b) contours of the regular shock reflection problem.

tively. At the initial time, the velocity is zero; the pressure and density are given as follows. (1) for the former, $(\rho/\rho_0, p/p_0) = (5, 5)$ when $\sqrt{(x-0.4)^2 + (y-0.4)^2 + (z-0.4)^2} \leq 0.3$ and others $(\rho/\rho_0, p/p_0) = (1, 1)$ with reflective boundary conditions at the surfaces; (2) for the latter, $(\rho/\rho_0, p/p_0) = (5, 5)$ when $\sqrt{(x-0.3)^2 + (y-0.7)^2 + (z-0.5)^2} \leq 0.2$ and others $(\rho/\rho_0, p/p_0) = (1, 1)$ with periodic boundary conditions at the surfaces. The computational domain is $[0, 1] \times [0, 1] \times [0, 1]$.

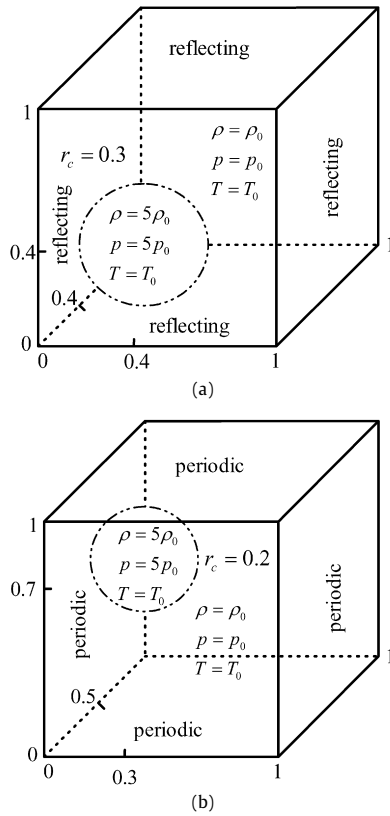


Fig. 4. Configurations of 3D explosion in a box (a) and 3D expanding shock in open space (b).

In our simulations, a uniform mesh $N_x \times N_y \times N_z = 120 \times 120 \times 120$ is used. The characteristic temperature T_c is set to be T_0 . The time step Δt is set to be $20000\tau_0$ and $10000\tau_0$ for the tests, respectively. The simulation of 3D explosion in a box takes 4 hours and 25 minutes on a computer with an Intel(R) Xeon(R) CPU E5410/2.33GHz. The density contour of 3D explosion in a box at $z = 0.4$ ($t = 0.5t_0$, $t_0 = L_0/u_0$ and $L_0 = 1$ m) is presented in Fig. 5, while the density contours of 3D expanding shock in open space at $z = 0$ and $z = 0.5$ ($t = 0.84t_0$) are presented in Figs. 6 and 7, respectively. For comparison, the results reported in Refs. [28,29], which were obtained via the block-structured adaptive mesh refinement technique and can be taken as benchmark solutions, are also shown in these figures. Excellent agreement can be found by comparing the present numerical results with those given in Refs. [28,29]. The complex features such as the shocks and the interactions between the shocks are well captured.

4. Conclusions

In this Letter, a 3D non-free-parameter LB model has been presented following the non-free-parameter LB method. The main features of the non-free-parameter LB method are retained in the present model. Furthermore, compared with the 2D non-free-parameter LB model shown in Ref. [16], the present model has some additional features: first, a recently developed approach, through which the specific-heat ratio can be easily adjusted, is incorporated into the model; second, an IMEX finite-difference scheme has been used to solve the discrete Boltzmann-BGK equation. Due to the excellent performance of the IMEX scheme, the numerical stability and the computational convergence rate are improved. Meanwhile, a relatively large time step can be chosen.

The model has been validated by its applications to some typical inviscid compressible flows ranging from 1D to 3D. The

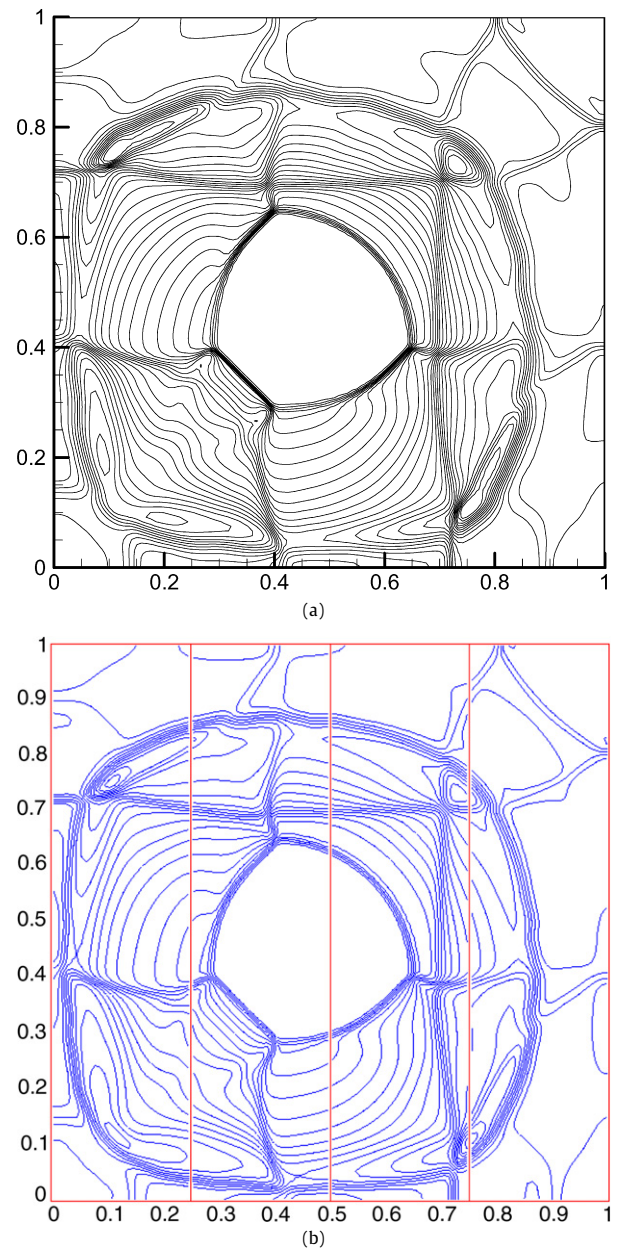
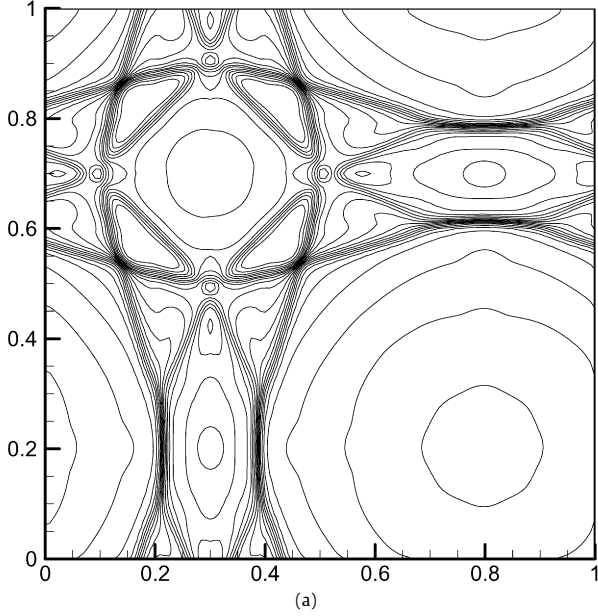


Fig. 5. Density contour of explosion in a box at $z = 0.4$. 30 equally spaced contour lines from $\rho/\rho_0 = 0.2$ to $\rho/\rho_0 = 2.3$. (a) present and (b) Ref. [28].

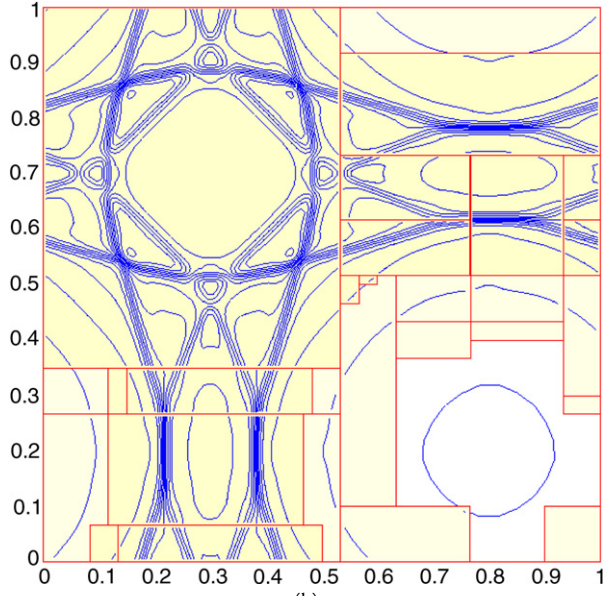
numerical results are found to be in excellent agreement with the analytical solutions and/or other numerical results. When the finite-difference LB scheme is used, there may be little difference between a microscopic/mesoscopic method and a macroscopic method such as the finite-difference method. Nevertheless, as pointed out by many researchers [10,11,16], the LB method, being simple in structure, utilizing linear operators, avoiding non-linear convective terms, and obviously parallelizable due to its explicit scheme and the local interaction, is worth further developing.

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(a)



(b)

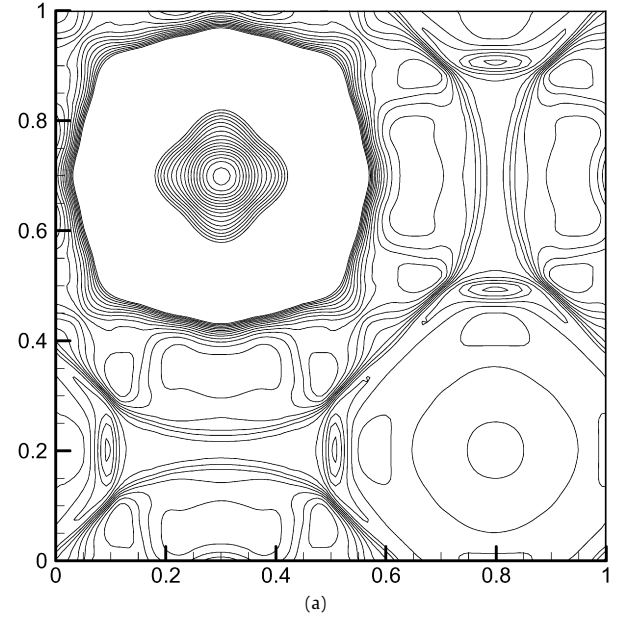
Fig. 6. Density contour of expanding shock in open space at $z = 0.24$ equally spaced contour lines from $\rho/\rho_0 = 0.7$ to $\rho/\rho_0 = 1.3$. (a) present and (b) Ref. [29].

Appendix A. Discrete equilibrium distribution function of the 3D non-free-parameter LB model

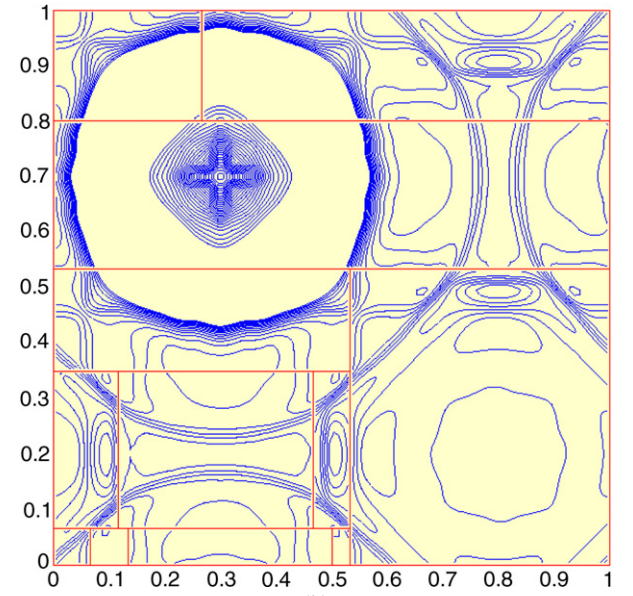
$$f_0^{eq} = \frac{1}{60} \rho (60 - 75\bar{r}^2 - 75\bar{u}^2 - 75\bar{v}^2 - 75\bar{w}^2 + 21\bar{r}^4 + 15\bar{u}^4 + 15\bar{v}^4 + 15\bar{w}^4 + 70\bar{u}^2\bar{r}^2 + 70\bar{v}^2\bar{r}^2 + 70\bar{w}^2\bar{r}^2 + 60\bar{u}^2\bar{v}^2 + 60\bar{u}^2\bar{w}^2 + 60\bar{v}^2\bar{w}^2),$$

$$f_1^{eq} = -\frac{1}{90} \rho (15\bar{u}^4 + 15\bar{u}^3 - 60\bar{u} + 45\bar{u}\bar{r}^2 + 45\bar{u}\bar{v}^2 + 45\bar{u}\bar{w}^2 + 45\bar{u}^2\bar{v}^2 + 45\bar{u}^2\bar{w}^2 + 60\bar{u}^2\bar{r}^2 + 15\bar{v}^2\bar{r}^2 + 15\bar{w}^2\bar{r}^2 - 60\bar{u}^2 + 9\bar{r}^4 - 20\bar{r}^2),$$

$$f_7^{eq} = \frac{1}{60} \rho (15\bar{u}^2\bar{v}^2 + 15\bar{u}^2\bar{v} + 15\bar{u}\bar{v}^2 + 15\bar{u}\bar{v} + 5\bar{u}^2\bar{r}^2 + 5\bar{v}^2\bar{r}^2 + 5\bar{u}\bar{r}^2 + 5\bar{v}\bar{r}^2 + \bar{r}^4),$$



(a)



(b)

Fig. 7. Density contour of expanding shock in open space at $z = 0.5$. 25 equally spaced contour lines from $\rho/\rho_0 = 0.9$ to $\rho/\rho_0 = 1.65$. (a) present and (b) Ref. [29].

$$f_{19}^{eq} = \frac{1}{360} \rho (15\bar{u}^4 + 30\bar{u}^3 + 30\bar{u}^2\bar{r}^2 + 30\bar{u}\bar{r}^2 - 15\bar{u}^2 - 30\bar{u} - 5\bar{r}^2 + 3\bar{r}^4),$$

$$\begin{aligned} f_1^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(\bar{u}, \bar{v}, \bar{w}), & f_2^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(-\bar{u}, \bar{v}, \bar{w}), \\ f_3^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(\bar{v}, \bar{u}, \bar{w}), & f_4^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(-\bar{v}, \bar{u}, \bar{w}), \\ f_5^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(\bar{w}, \bar{v}, \bar{u}), & f_6^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_1^{eq}(-\bar{w}, \bar{v}, \bar{u}), \\ f_7^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{u}, \bar{v}, \bar{w}), & f_8^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{u}, -\bar{v}, \bar{w}), \\ f_9^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{u}, -\bar{v}, \bar{w}), & f_{10}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{u}, \bar{v}, \bar{w}), \\ f_{11}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{u}, \bar{w}, \bar{v}), & f_{12}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{u}, -\bar{w}, \bar{v}), \\ f_{13}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{u}, -\bar{w}, \bar{v}), & f_{14}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{u}, \bar{w}, \bar{v}), \\ f_{15}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{v}, \bar{w}, \bar{u}), & f_{16}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{v}, -\bar{w}, \bar{u}), \\ f_{17}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(\bar{v}, -\bar{w}, \bar{u}), & f_{18}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_7^{eq}(-\bar{v}, \bar{w}, \bar{u}), \end{aligned}$$

$$\begin{aligned}
f_{19}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(\bar{u}, \bar{v}, \bar{w}), & f_{20}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(-\bar{u}, \bar{v}, \bar{w}), \\
f_{21}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(\bar{v}, \bar{u}, \bar{w}), & f_{22}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(-\bar{v}, \bar{u}, \bar{w}), \\
f_{23}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(\bar{w}, \bar{v}, \bar{u}), & f_{24}^{eq}(\bar{u}, \bar{v}, \bar{w}) &= f_{19}^{eq}(-\bar{w}, \bar{v}, \bar{u}),
\end{aligned} \tag{A.1}$$

where $\bar{u} = u_x/c$, $\bar{v} = u_y/c$, $\bar{w} = u_z/c$, and $\bar{r} = r/c = \sqrt{T/T_c}$. Here $f_3^{eq}(\bar{u}, \bar{v}, \bar{w}) = f_1^{eq}(\bar{v}, \bar{u}, \bar{w})$ means $f_3^{eq}(\bar{u}, \bar{v}, \bar{w})$ is calculated with the same form of $f_1^{eq}(\bar{u}, \bar{v}, \bar{w})$, but the locations of $\bar{u}$ and $\bar{v}$ in $f_1^{eq}(\bar{u}, \bar{v}, \bar{w})$ are permuted.

Appendix B. Chapman–Enskog analysis

By introducing the multi-scale expansions, we expand the space derivative, time derivative, and the distribution function, respectively, as

$$\nabla = K \nabla_1, \tag{B.1a}$$

$$\partial_t = K \partial_{t1} + K^2 \partial_{t2} + \dots, \tag{B.1b}$$

$$f_i = f_i^{eq} + K f_i^{(1)} + K^2 f_i^{(2)} + \dots, \tag{B.1c}$$

where K is a small parameter proportional to the Knudsen number. When the macroscopic variables are calculated as follows:

$$\begin{aligned}
\rho &= \sum_i f_i, & \rho \mathbf{u} &= \sum_i f_i \mathbf{e}_i, \\
\rho \frac{1}{2} (u^2 + DRT) &= \sum_i f_i \frac{e_i^2}{2},
\end{aligned} \tag{B.2}$$

$f_i^{(n)}$ would be constrained by

$$\begin{aligned}
\sum_i f_i^{(n)} &= 0, & \sum_i f_i^{(n)} \mathbf{e}_i &= 0, \\
\sum_i f_i^{(n)} e_i^2 &= 0, & n &= 1, 2, \dots
\end{aligned} \tag{B.3}$$

Substituting Eqs. (B.1) into Eq. (11), we can obtain the equations for the first- and second-order expansions in K , respectively.

$$(\partial_{t1} + \mathbf{e}_\alpha \cdot \nabla_1) f_i^{eq} + \frac{1}{\tau} f_i^{(1)} = 0, \tag{B.4}$$

$$\partial_{t2} f_i^{eq} + (\partial_{t1} + \mathbf{e}_\alpha \cdot \nabla_1) f_i^{(1)} + \frac{1}{\tau} f_i^{(2)} = 0. \tag{B.5}$$

Taking the zeroth-order through second-order moments of Eq. (B.4) and using Eqs. (3a), (3b), and (3d), we have

$$\partial_{t1} \rho + \nabla_1 \cdot (\rho \mathbf{u}) = 0, \tag{B.6}$$

$$\partial_{t1} (\rho u_j) + \partial_{1\alpha} (\rho u_\alpha u_\beta) = -\partial_{1\beta} p, \tag{B.7}$$

$$\partial_{t1} (\rho E) + \partial_{1\gamma} [(\rho E + p) u_\gamma] = 0. \tag{B.8}$$

Similarly, taking the zeroth-order and the first-order moments of Eq. (B.5), we can obtain the following equations:

$$\partial_{t2} \rho = 0, \tag{B.9}$$

$$\partial_{t2} (\rho u_\beta) + \partial_{1\alpha} \left(\sum_i e_{i\alpha} e_{i\beta} f_i^{(1)} \right) = 0. \tag{B.10}$$

With the aid of Eq. (B.4), $\sum_i e_{i\alpha} e_{i\beta} f_i^{(1)}$ can be written as

$$\begin{aligned}
&\sum_i e_{i\alpha} e_{i\beta} f_i^{(1)} \\
&= -\tau \left[\partial_{t1} \left(\sum_i e_{i\alpha} e_{i\beta} f_i^{eq} \right) + \partial_{1\gamma} \left(\sum_i e_{i\alpha} e_{i\beta} e_{i\gamma} f_i^{eq} \right) \right].
\end{aligned} \tag{B.11}$$

Combining Eq. (B.6) with Eq. (B.7), after some standard algebra, we can obtain

$$\partial_{t1} (\rho u_\alpha u_\beta) = -u_\alpha \frac{\partial p}{\partial r_{1\beta}} - u_\beta \frac{\partial p}{\partial r_{1\alpha}} - \frac{\partial}{\partial r_{1\lambda}} (\rho u_\alpha u_\beta u_\gamma). \tag{B.12}$$

Combining Eq. (B.8) with Eq. (B.7), we can obtain

$$\partial_{t1} p = -\partial_{1\gamma} (p u_\gamma) - \frac{2}{D} p \partial_{1\gamma} u_\gamma. \tag{B.13}$$

Using Eqs. (3c), (3b), (B.12), and (B.13), we can simplify Eq. (B.11) as

$$\sum_i e_{i\alpha} e_{i\beta} f_i^{(1)} = -\tau p \left(\frac{\partial u_\alpha}{\partial r_{1\beta}} + \frac{\partial u_\beta}{\partial r_{1\alpha}} - \frac{2}{D} \frac{\partial u_\gamma}{\partial r_{1\gamma}} \delta_{\alpha\beta} \right). \tag{B.14}$$

Adding Eq. (B.6) and K times Eq. (B.9) yields the continuity equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0. \tag{B.15}$$

Combining Eqs. (B.7) and (B.10) with Eq. (B.14), we can obtain the momentum equation

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \Pi + O(K^2). \tag{B.16}$$

The viscous stress tensor Π is given by

$$\Pi = \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{D} (\nabla \cdot \mathbf{u}) \mathbf{I} \right], \tag{B.17}$$

where $\mu = \tau p$ is the shear viscosity and $\mathbf{I}$ is the unit tensor. Finally, Eq. (B.8) gives the energy equation at the Euler level

$$\partial_t (\rho E) + \nabla \cdot [(\rho E + p) \mathbf{u}] = 0 + O(K). \tag{B.18}$$

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